EIE3123 DYNAMIC ELECTRONIC SYSTEMS: QUIZ 5

1. A unity feedback system has an open-loop transfer function given by

$$G(s)=\frac{250}{s[0.1s+1]}$$

Design a lag compensator added in series with the plant to yield a phase margin of 50°. Use the matched pole-zero approximation, determine an equivalent digital realization of the compensator. Write down the resulting difference equation for implementing the lag compensator in a microcontroller.

2. A machine-tool system has the form

$$G(s)=\frac{0.1}{s[s+0.1]}$$

The sampling rate is $T = 1$ s. It is desired to achieve a step response with 16% overshoot and a settling time of 12 seconds. Design a compensator directly in the z-domain to achieve these specifications. Write down the resulting difference equation for implementing the compensator in a microcontroller.